

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

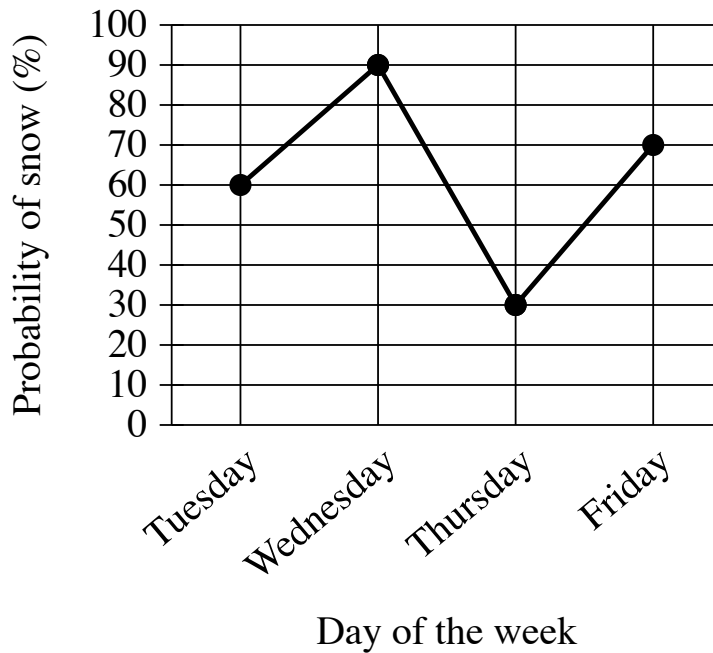
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# Two-variable data: Models and scatterplots

# 01

10 questions · ~15 min





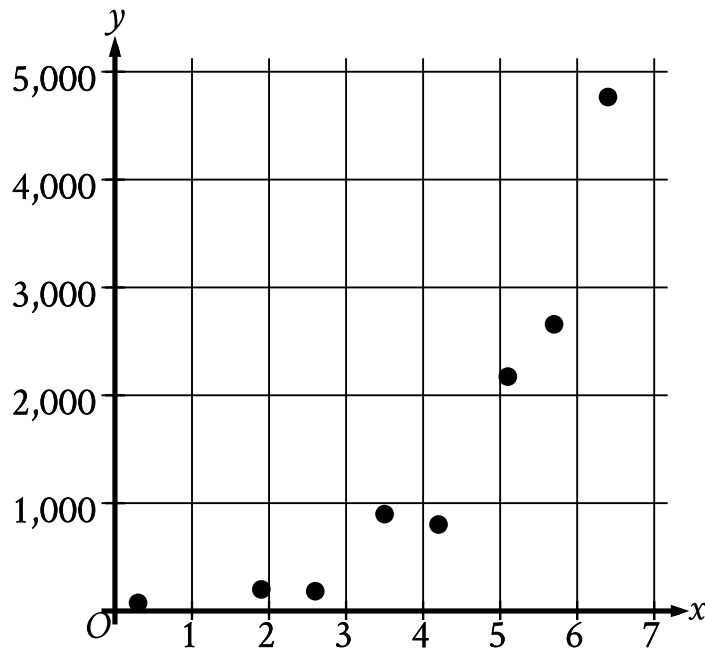
- The line graph:
  - Begins at Tuesday, 60%
  - Rises sharply to Wednesday, 90%
  - Falls sharply to Thursday, 30%
  - Rises sharply to Friday, 70%

The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow 30%?

- A. Tuesday
- B. Wednesday
- C. Thursday

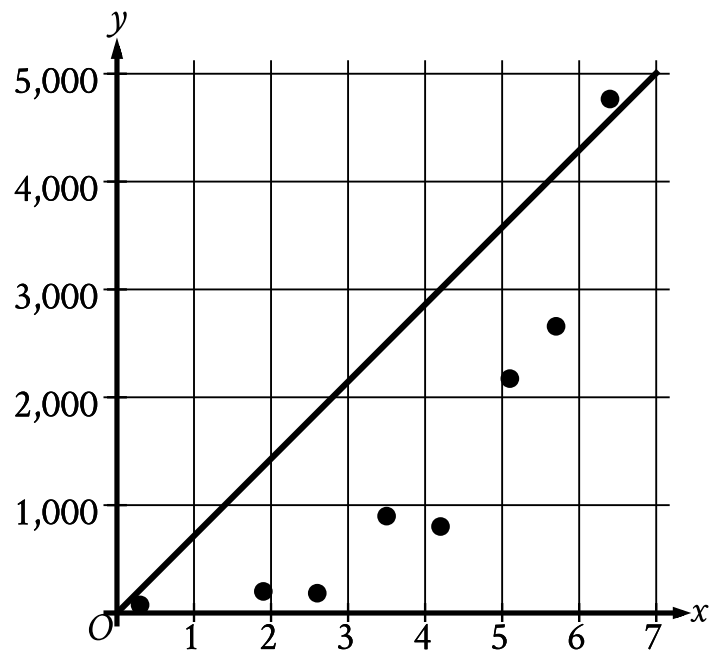
**D. Friday**

The scatterplot shows the relationship between two variables,  $x$  and  $y$ .



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:
  - (0.3 comma 75)
  - (1.9 comma 200)
  - (2.6 comma 183)
  - (3.5 comma 898)
  - (4.2 comma 802)
  - (5.1 comma 2,173)
  - (5.7 comma 2,658)
  - (6.4 comma 4,766)

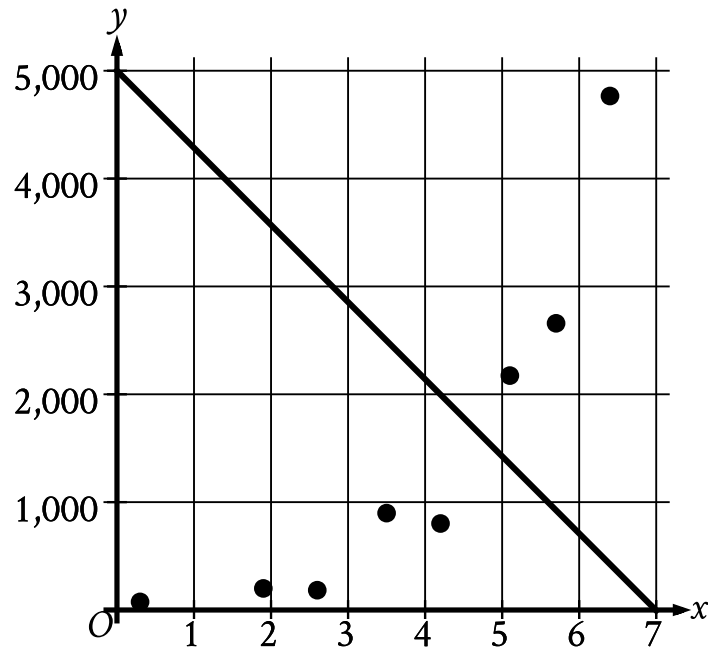
Which of the following graphs shows the most appropriate model for the data?



A.

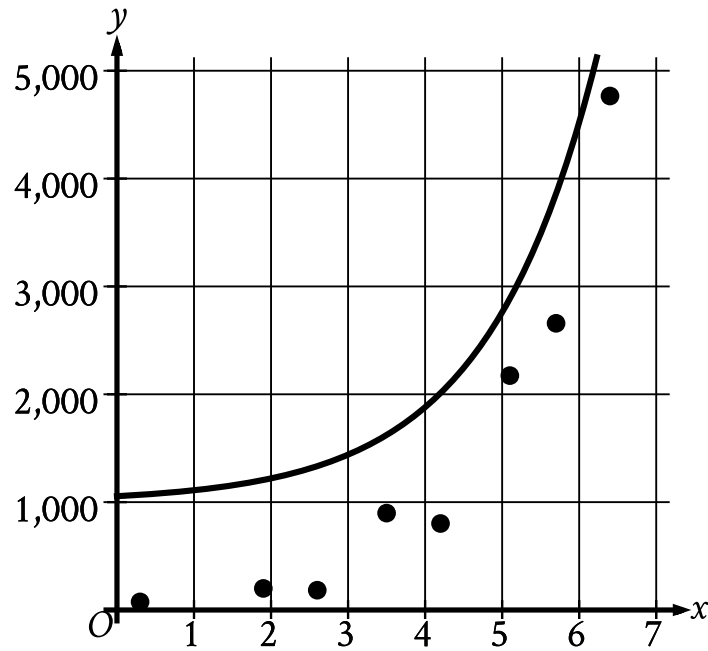
- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - 0 points are touching the line of best fit.
  - 1 point is above the line of best fit.
  - 7 points are below the line of best fit.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 0)
    - (6 comma 4,290)

B.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - 0 points are touching the line of best fit.
  - 3 points are above the line of best fit.
  - 5 points are below the line of best fit.
  - The line of best fit passes through the following coordinates:
    - (0 comma 5,000)
    - (6 comma 710)

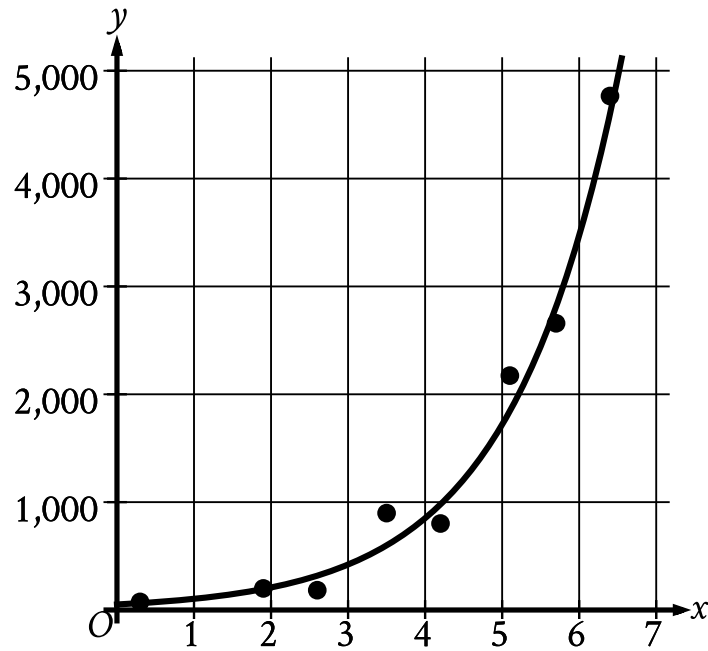
c.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
  - The curve of best fit trends up from left to right.
  - 0 points are touching the curve of best fit.
  - 0 points are above the curve of best fit.
  - 8 points are below the curve of best fit.
  - The curve of best fit passes through the following coordinates:
    - (0 comma 1,055)
    - (4 comma 1,880)
    - (6 comma 4,520)

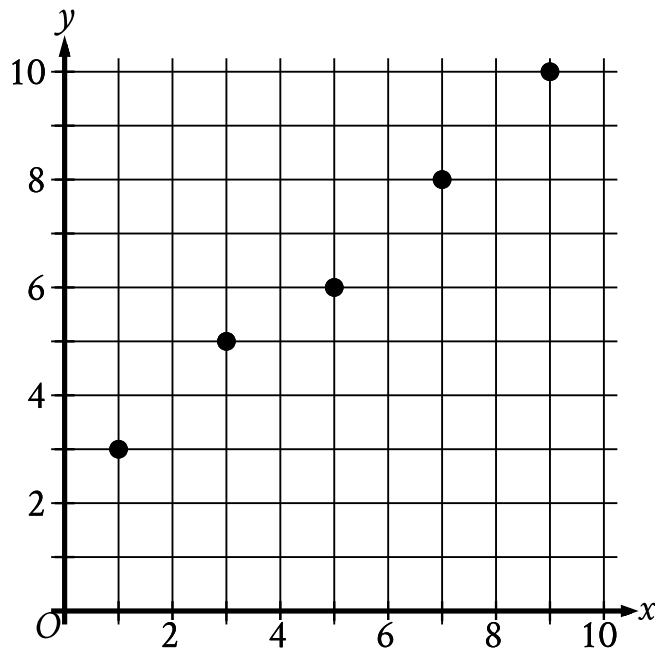


D.



- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
  - The curve of best fit trends up from left to right.
  - 2 points are touching the curve of best fit.
  - 3 points are above the curve of best fit.
  - 3 points are below the curve of best fit.
  - The curve of best fit passes through the following approximate coordinates:
    - (0 comma 51)
    - (4 comma 851)
    - (6 comma 3,472)

The scatterplot shows the relationship between two variables,  $x$  and  $y$ .



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
  - (1 comma 3.0)
  - (3 comma 5.0)
  - (5 comma 6.0)
  - (7 comma 8.0)
  - (9 comma 10.0)

Which equation is the most appropriate linear model for this relationship?

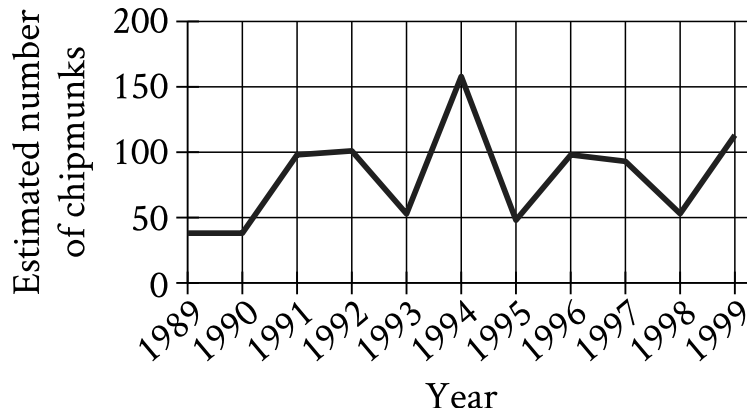
A.  $y = -0.9x - 2.2$

B.  $y = -0.9x + 2.2$

**C.**  $y = -0.9x$

**D.**  $y = 0.9x + 2.2$

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



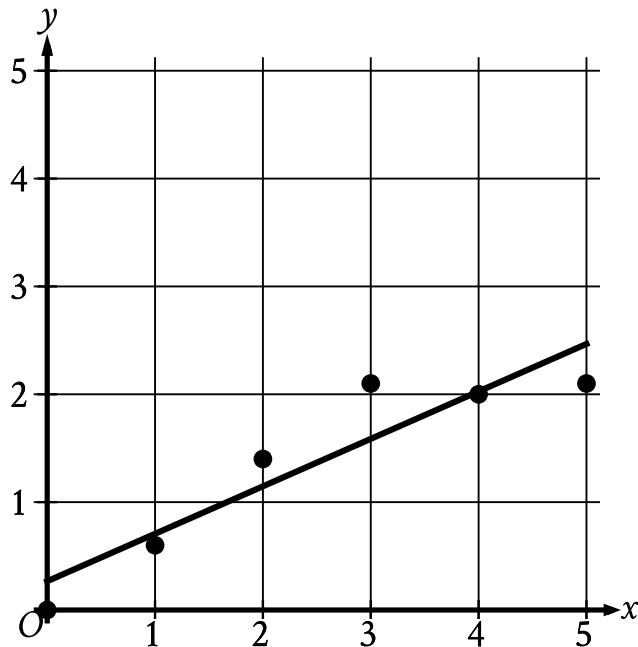
- The line graph:
  - Begins at 1989, 38 chipmunks
  - Remains level to 1990, 38 chipmunks
  - Rises sharply to 1991, 98 chipmunks
  - Rises gradually to 1992, 101 chipmunks
  - Falls sharply to 1993, 53 chipmunks
  - Rises sharply to 1994, 158 chipmunks
  - Falls sharply to 1995, 48 chipmunks
  - Rises sharply to 1996, 98 chipmunks
  - Falls gradually to 1997, 93 chipmunks
  - Falls sharply to 1998, 53 chipmunks
  - Rises sharply to 1999, 113 chipmunks

Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

- A. 1989
- B. 1994
- C. 1995

D. 1998

The scatterplot shows the relationship between  $x$  and  $y$ . A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - 1 point is touching the line of best fit.
  - 2 points are above the line of best fit.
  - 3 points are below the line of best fit.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 0.2)
    - (4 comma 2)

Which of the following is closest to the slope of the line of best fit shown?

A. -2.27

B. -0.44

C. 0.44

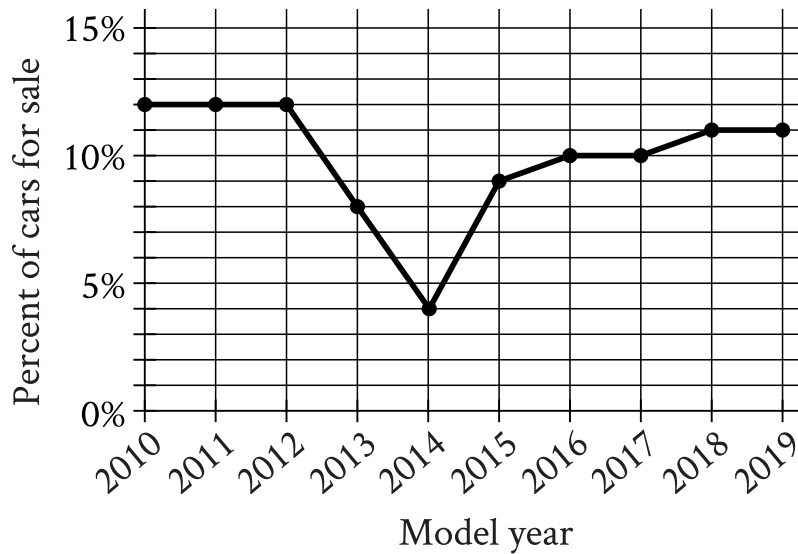
D. 2.27

An airplane descends from an altitude of 9,500 feet to 5,000 feet at a constant rate of 400 feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear



The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



- The line graph:
  - Begins at 2010, 12%
  - Remains level to 2011, 12%
  - Remains level to 2012, 12%
  - Falls sharply to 2013, 8%
  - Falls sharply to 2014, 4%
  - Rises sharply to 2015, 9%
  - Rises gradually to 2016, 10%
  - Remains level to 2017, 10%
  - Rises gradually to 2018, 11%
  - Remains level to 2019, 11%

For what model year is the percent of cars for sale the smallest?

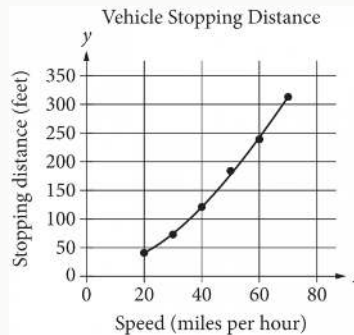
A. 2012

B. 2013

**C.** 2014

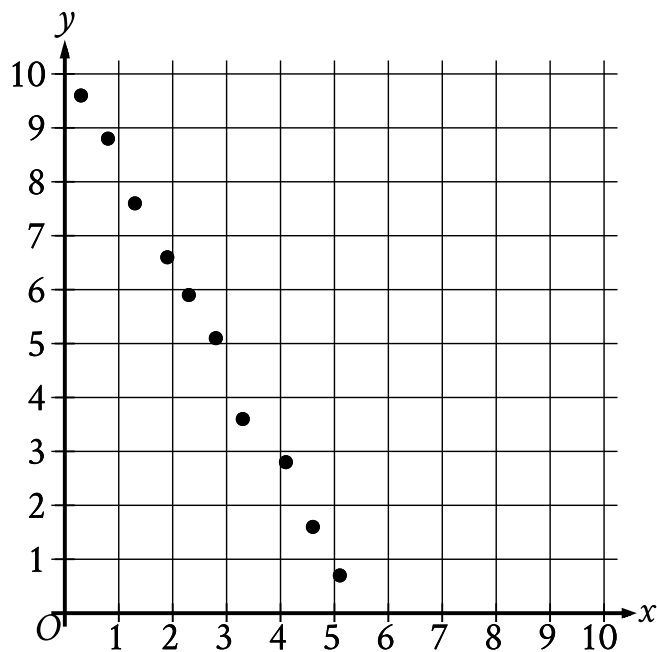
**D.** 2015

A study was done to determine a new car's stopping distance when it was traveling at different speeds. The study was done on a dry road with good surface conditions. The results are shown below, along with the graph of a quadratic function that models the data.



According to the model, which of the following is the best estimate for the stopping distance, in feet, if the vehicle was traveling 55 miles per hour?

- A. 25
- B. 30
- C. 210
- D. 250



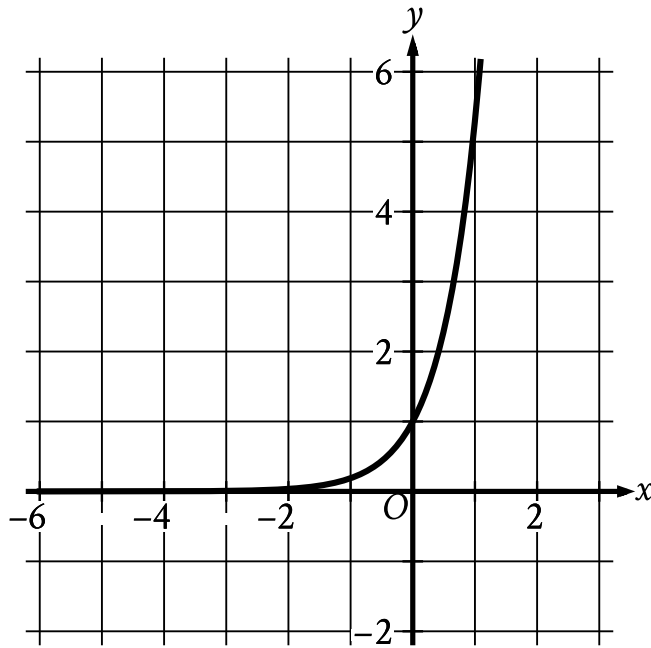
- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- Select data points have the following approximate coordinates:
  - (0.8 comma 8.9)
  - (2.8 comma 5.1)
  - (4.1 comma 2.9)

Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A.  $y = -1.9x - 10.1$
- B.  $y = -1.9x + 10.1$
- C.  $y = 1.9x - 10.1$

**D.**  $y = 1.9x + 10.1$

The graph of the function  $f$  is shown, where  $y = fx$ .



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 1).
  - In quadrant 1, the curve trends up sharply.
- As  $x$  decreases, the curve approaches the line  $y$  equals 0.
- The curve passes through the following points:
  - approximately (negative 1 comma 0.2)
  - (0 comma 1)
  - approximately (1 comma 5.3)

Which of the following best describes the function  $f$ ?

A. Decreasing exponential

- B.** Increasing exponential
- C.** Decreasing linear
- D.** Increasing linear